



Exam No: 09

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Essay title: “Critically evaluate the continuing relevance of classical military strategic thought – from Sun Tzu to Mahan – in understanding contemporary defence and security challenges.”

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I hereby confirm that this essay is my unassisted work, has never been submitted for publication or assessment elsewhere and that all sources used in its composition have been cited.

Signed by Student

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Glossary

ASAT	Anti Satellite
ELINT	Electronic Intelligence
ESA	European Space Agency
EW	Electronic Warfare
DSPS	Defence Support Program Satellites
GEO	Geostationary Earth Orbit
GEOINT	Geospatial Intelligence
GNSS	Global Navigation Satellite Systems
HEO	High Earth Orbit
ICBM	Intercontinental Ballistic Missile
IMINT	Imagery Intelligence
LEO	Low Earth Orbit
LOS	Lines of Communication
MEO	Medium Earth Orbit
MEV	Mission Extension Vehicle
NPT	Navigation, Precision and Timing
OODA	Observation, Orientation, Decision, Action
OST	Outer Space Treaty
RORSAT	Radar Ocean Reconnaissance Satellites
SATCOM	Satellite Communications
SDA	Space Domain Awareness
SIGINT	Signals Intelligence

Krieg ist die Fortführung der Politik, mit orbitalem Bombardement¹.

“In the bluntest of statements: space warfare is a certainty in the future because the use of space in war has become vital” (Gray, p. 307 2005). Today's great powers have fielded teams in the space arena since the 1950s, the fluidity of today's arena, emergence of new teams, untested primacy of old teams, and twenty first century field play has set the space arena for a new episode of *Star Wars: The Continuation Wars*.

The modern utility of space capabilities derives from four decades of securitisation (Bowen, 2021), the past two decades have seen military systems transition to civilian application, from submarine tracking to agriculture tracking, cruise missile navigation to traffic navigation (Muraviev, 2021). From a military perspective space infrastructure supports tactical battlefield applications, precision bombing, networked communication systems, long range drone operations and intelligence collection. Modern states depend on space services for societal functions such as financial transactions, electricity grid performance, smartphones, aircraft and shipping management, etc., a denial of service would result in significant social and economic disruption (MOD, 2022; USSF 2025).

Carl Von Clausewitz maxim, *“War is a continuation of policy, by other means”*, provides a possible military strategy framework to analysis and translate the ways of space dynamics, this paper synthesises key aspects of Clausewitz's military war theory, offers a foundation for strategic thinking and its relevant application to the space domain and security politics. The paper orbits around Clausewitz's groundwork on the trinity, centre of gravity, fog, friction, probability of chance and the operating environment, assessing their relevance, dynamics and relationship to the space domain. The paper critically evaluates the application of classical theory to contemporary practice within the space security architecture.

Julian Corbett's maritime works *Some Principles* are used to critique Clausewitz theories, to gain an understanding within a separate domain, transferable and relevance to the modern age, they offer a nuanced understanding of how strategy bridges the gap from theory to practice.

Trinity

Clausewitz capstone doctrine *Dreifaltigkeit*, the “trinity”, synthesises the nature of war and argues that to understand war an analysis on all three forces is required. These forces comprise of: primordial

¹ War is the continuation of politics, with orbital bombardment.

violence, hatred and enmity – the people; the play of chance or probability - the military commanders skill and talent - relatable to military leadership, capabilities and readiness in a contemporary setting; and wars element as an instrument to achieve political aims – the government (Bassford, 2023). Clausewitz explains that the trinity floats between all three dimensions, best represented and viewed like the deterministic dynamics on a randomly oscillating magnetic pendulum, when one force is altered it effects the equilibrium of the pendulum.

The proximity between all three forces was evident during the *limited* wars of Vietnam, Iraq and Afghanistan with events on the battlefield having emotional impacts on the people and political leadership, which in turn affected military capabilities when resources or policy was restrained (Bassford, 2023). Clausewitz concept of *total war* was applied by Bernard Brodie in his theories on nuclear warfare and deterrence, that the threat and feasibility of total destruction by nuclear weapons created a credible deterrence. But within the space environment, where actions maybe invisible and mostly intangible, despite a hyper reliance on space infrastructure, does this provide an opportunity to conduct war by play of chance without consequence, or does it required significant domain security to prevent chaos.

The securitisation of space presents challenges to government leaders, military commanders, and the people. The United States Space Force capstone publication defines the space environment as “*contested, congested, and competitive*” (USSF, 2025), with the UK Space doctrine adding “*complexity*” to the equation (MOD, 2025), signifying the relationship between civil and military, or government and commercial entities.

Competition between great power rivals has occurred below the threshold of direct conflict, due to nuclear weapons (Shapiro et al., 2024), the multi-domain grey space is where adversaries now operate, testing system resilience, or jostling for strategic leadership roles (USSF, 2025; MOD, 2022). Military space forces currently play a critical role in armed conflict, either through the application of force, or as a strategic enabler in other domains.

A synthesis of mission statements from Russian Space Forces, UK Space Command and the United States Space Force (USSF) outlines similar mission verbs, “protect, defend, warning, and secure” also defining areas of responsibilities, “*UK, allied interests, sovereignty (Russia), Nation (US)*”. The USSF doctrine outlines method through strategic and operational approaches, and effects of deterrence, compel, deny, by kinetic and non-kinetic action (USSF, 2025). The mission statements outline the demanding tasks and vast area of responsibilities, and the requirement for a comprehensive approach of political, civil and military to space security .

Clausewitz trinity’s to space outlines that warfare is not just mechanical, connectivity and information shape human emotion and political decisions. A denial of service by an adversary, could start a burn

on a short emotional human fuse, destabilising the trinity and setting political leaders for confrontation.

Centre of Gravity

Clausewitz's operational design is based around the *Schwerpunkt*² or Centre of Gravity (COG) concept, other design aspects such as objectives, lines of operations (LOO), line of effort (LOE) and termination state being components within the planning process. The concept of the operational art, is to establish the means and ways, to mass effects against adversaries *source of power*, gain and maintain relative advantage, to reach the military desired end state. Contemporary teaching and application of the COG enables commanders and staff to visualise and describe the operational approach, to achieve objectives, and ensure mission accomplishment (JDP 3-0, 2017).

Clausewitz's COG is understood as the source of power, where gravitational forces converge within an object, with the objects weight balanced in all directions. Clausewitz refers that the COG "*is against that part of the enemy's forces, where they are most concentrated that, if a blow were to occur, the effect would emanate the furthest; furthermore, the greater the mass our own forces possess when they deliver the blow, the more certain we can be of the blow's success*" (Clausewitz, p.485 1832). The COG concept is only applicable as a warfighting effort when both *Einheit and Zusammenhang* (unity and interdependence or connectivity) is present between the adversaries force and the space they occupy (Clausewitz, 1832), enabling numerous adversarial COGs on multiple domains (CADD, 2021).

US military service branches have shaped their COG definition through the optic of *Einheit und Zusammenhang*, reflecting their services unique function, characteristics and capabilities. The USMC a small sized force designed to win battles rather than enduring long protracted wars, define the adversarial COG as "*any important sources of strength*" (USN, 1997), while the US Army, a much larger force capable of conducting large scale campaigns identifies the adversaries COG as the "source of strength" (U.S. Army, 2022), outlining a much broader and deeper definition. The Air Force views COGs as "*strategic and operational critical points*" (USAF, 2020), reflecting the role and application of air power in a campaign. The US Joint Staff has diverged from Clausewitz effects based concept of a COG, and applies it through a capabilities optic, "*characteristics, capabilities, or locations from which military forces derives its freedoms of action, physical strength, or will to fight,*"

² Schwerpunkt is the term used most frequently; it appears some fifty-three times in original German *On War*.

(JP 3-0, 2017). As the supreme definition within the military nesting, a number of deductions can be made to shape a space based concept for COG analysis.

The five other operating domains³ are interdependent and relevant to space power, therefore each domain must be assessed individually to establish a relevant *Schwerpunkt*, that focus on supporting space systems and their infrastructure. Within the land domain arguable the most critical COG for space power, it encompasses ground stations, control centres, personnel, launch facilities and infrastructure. The space domain COGs requires systems that provide space situational awareness (SSA) and space surveillance and tracking (SST) which enable SDA, mission essential for protecting own systems and targeting adversary's (MOD, 2022; Sheldon, 2021). Next the cyber domain, which includes hardware and software infrastructure connecting space systems to civil and military end users. Followed by both the maritime and air domains, with tracking and communication assets and access area denial (A2D) systems, with the latter being extremely time dependent due to the nature of aircraft. The Electronic Warfare (EW) domain involves the use of electromagnetic energy providing a *Zusammenhang* capability from land, air, maritime and cyber to the space domain. From a multi-domain operation, EW is also key for enabling space based operations.

Fog and Friction

Clausewitz wrote, *“Everything in war is very simple, but the simplest thing is difficult. The difficulties accumulate and end by producing a kind of friction”*. No unified definition or international law exists to define where space begins, and earth's atmosphere ends, but physics provides the distinction - where laws of orbital mechanics direct the movement of objects in comparison to aerodynamics (MOD, 2022; Muraviev, 2021). The absence of a clear demarcation creates a Rumsfeld⁴ logic of a 'known unknown' or Clausewitzian *fog*, it affects national sovereignty claims over airspace, overflights of high altitude aircraft, balloons and space objects - which permits reconnaissance without violating laws.

Intelligence as a source of Fog.

Clausewitz states in *Intelligence in War* (Clausewitz, p. 117 1832) that information pertains to *“all the knowledge which we have of the enemy and his country”* and is the foundation which all decision and concepts are based. Space operations require systems that provide space situational awareness

³ NATO and Irish military doctrine outline six operating domains: land, maritime, air, space, cyber and electronic warfare.

⁴ On 12th February, 2002 Donald Rumsfeld U.S. SECDEF outlined his concept of fog, famed as the 'known knowns' media brief.

(SSA), and space surveillance and tracking (SST), to developing Space Domain Awareness (MOD, 2025; Muraviev, 2021) to enable effective decision making.

Highly critical of intelligence, stating it's contradictory nature, slamming its reliability, accuracy and credibility, Clausewitz argues that an officer's comprehensive knowledge and effective judgment can overcome such folly, making intelligence based decisions irrelevant.

Clausewitz writings on information must be given context, where the passage of information took hours, days, or weeks to reach decision makers, then likely outdated, and therefore irrelevant. In today's global network of real time connectivity, with improved systems and analysis, information is mission critical in all levels of war and in all domains.

Space surveillance tracks objects in or entering the space, from a network of ground based radars, telescopes, sensors and interpreted by robust data analysis, delivering effective SSA (MOD, 2022; Sheldon, 2021). Reverting to Clausewitz demand for commanders to hold comprehensive understanding of the environment and resources, SSA requires sufficient understanding of risks, hazards and complexities within the domain. The fusion of SST, SSA and security focused intelligence provides SDA (MOD, 2022).

Despite a system of systems providing information, a Clausewitzian "*fog*" lingers with the intelligence reliability on the actual capabilities and purpose of space objects, or the Rumsfeld known unknown logic of space objects. Observing through the 'fog' is the function of military intelligence, a task made more complicated by other external factors. Ascertaining the true purpose and intent of space systems presents challenges: current international registration only requiring basic information (Sheldon, 2021); further complication with dual use capabilities create '*unknown unknowns*' conundrums for commanders and political leaders. In 1985 Soviet leaders believed that the change in mission parameters for a U.S. space shuttles launch was to capture the uncrewed and unresponsive orbiting Saylut-7 space station (Hendrickx, 2014; Bowen, 2021).

Adversarial satellites can remain concealed along congested LOC amongst space debris, the application of electromagnetic stealth materials on payloads⁵ renders them undetectable (Zhao et al, 2025), all these complexities cloak the intelligence space in fog.

Advances in robotics and communication equipment, three dimensional printing and the ability to cannibalise obsolete satellites parts has created a new dimension for on-orbiting maintenance by mission extension vehicles (MEV) (McKeon, 2025). The dual use of an adversaries MEV could render friendly satellites inoperable. The inability for satellites to detect proximity threats, and further

⁵ Payload refers to the elements or parts of the spacecraft that perform the desired functions of the space object.

complicated by legal entanglement of a satellite true purpose provide a hybrid means for space warfare.

Clausewitz scepticism aligns closely with the challenges of space intelligence today, an overreliance on unrefined data from sources requires analysis, judgement and context interpretation.

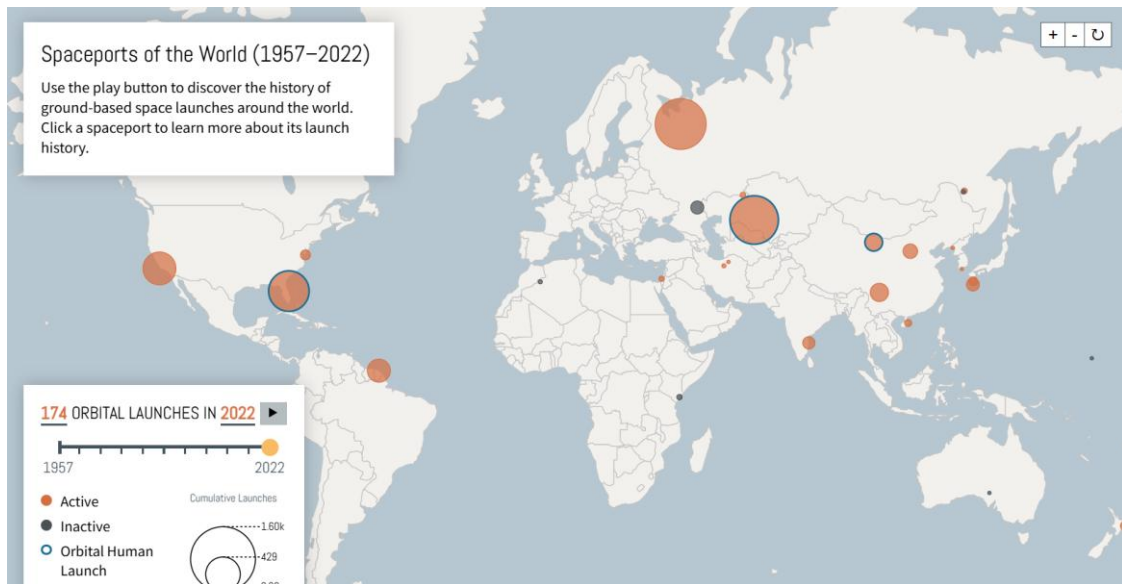
Weather as a source of friction.

Clausewitz introduction of *Friktion*, a believe that separates real war from paper war, he cites weather as both a source of friction, based on the element of chance - due to its unpredictability, and the opportunity it offers and denies to military forces. The space environment is also susceptible to weather: exploding stars release high-energy particles; and solar winds emitted from sun's corona create dynamic electromagnetic energy flows. The high-energy bombards space vehicles, degrading their efficiency and lifespan, while solar flares reduce, disrupt or block radio waves between ground based communications facilities and orbital vehicles. These astrological phenomena have also activated onboard sensor creating false readings, with a notable event occurring on 26 September 1983 – nearly triggering a Soviet nuclear response, based on *fog* and paranoia from the coinciding NATO Able Archer exercise (Cimbala, 2007). The most extreme space weather events - coronal mass ejections, can render sensitive earth based systems inoperable. Atmosphere drag, gravitational fluctuations and pull of other planetary bodies effects space objects orbits. Like terrestrial weather, space weather is hard to predict, its relationship is correlated to the eleven year activity cycle of the sun, the correcting for environmental factors requires manoeuvring space systems. Understanding of space weather is critical to space domain awareness (SDA) (USAF, 2020).

Predictability as Friction.

Clausewitz outlines that *chance and adversarial actions* cause military plans to diverge from the original plan, *leading to considerable friction*, and if not understood can dramatically alter the course of war. Paradoxically it creates opportunities for exploitation by commanders, if they poses the technical knowledge of their own and adversaries capabilities, vulnerabilities, and weaknesses. The predictability known launch points, orbit parameters, precise position, navigation and timing (PNT) of space objects creates vulnerable patterns for observing adversaries, enabling the generation of robust kill chain⁶. The illustration below displays primary space launch facilities, this 'known known' enables hostile actors to monitor, survey, sabotage or target ground facilities or space vehicles.

⁶ An integrated end-to-end process, described as Find, Fix, Track, Target, Engage, Assess (F2T2EA).



(Predictability - Spaceports of the World. Roberts, 2023)

The legal environment as a source of friction.

“War is not merely a political act, but a real political instrument, a continuation of political intercourse, to carry out the same by other means”, (Clausewitz, p. 280 1832). Friction is created by political acts or rules establishment. International treaties, national laws governing space use, ownership of objects, weapons platforms, nuclear power and battlespace management creates terrestrial tensions, further complicated with dual-use technologies.

In accordance with Outer Space Treaty (OTS) obligation, space is *“free for exploration and use by all States”*. The OTS recognises the supremacy of international law and the UN Charter, governing the right to use force, it prohibits the deployment of nuclear weapons in space, but provides legal freedom for objects transiting through space such as ICBM or SLBM

Transnational aerospace manufacturing and product licencing define organisation ownership, accountability and liability however is more vague (Sheldon, 2021). National laws provide regulatory frameworks, particularly in countries with rules based order, some aerospace companies have exploited regulatory weaknesses for commercial advantage. Swarm Technologies a U.S. registered company used legal loopholes to launch its satellites utilising an Indian aerospace company when denied a licence by the US Federal Aviation Authority (Vance, 2018). Small scale on, it outlines how national laws can be circumvented, but still internationally legal. Government owned space systems developed for civil applications such as satellite communication (SATCOM), weather observation, or pollution monitoring can be adapted or tasked for military needs.

Nuclear power in space is another a source of legal friction between states, Soviet radar ocean reconnaissance satellites (RORSATS) contained miniature nuclear reactors, as a *high value target*⁷ its destruction creates significant hazardous threat, similar to Kosmos 954 disintegration over Canada. The absence of legal definition to where space begins creates a rolling friction, additionally the ability mask a space systems full capabilities, intent and purpose through liability ambiguity, the use of transnational organisation, places them within an international legal grey zone (Muraviev, 2021).

Operating Environment

Clausewitz's chapters *Marches* (Clausewitz, p. 314 1832) and *On Physical Effort* (Clausewitz, p. 115 1832) are relatable to understand the environment and the required care and effectiveness of resources during operation - stating that precision movements are fundamental to conserve organisation resources. The single operational critical factor of the space environment are manoeuvrability limits of orbital systems. Space objects encounter drag, requiring minute orbital adjustments thrusts with chemical fuel, a design consideration factored during development. The operational lifespan of space systems is reducing by increased fuel consumption for delta-v⁸ manoeuvres to avoid space debris, a factor examined later in this paper. The physical dimension to space outlines the requirements for care of resources and an understanding of environment, the limitations and opportunities it possesses.

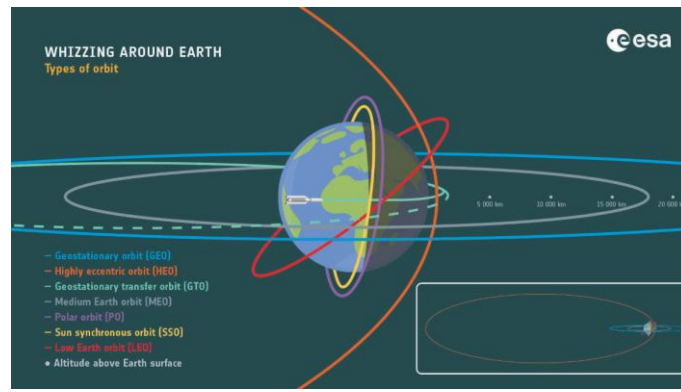
Lines of Communication

Clausewitz outlines his significant factors for lines of communication (LOC): *length*; *number*; *situation*; *nature*; and *protection* they derive (Clausewitz, p. 345 1832) which in present day hold equal meaning with the space domain.

The *length* directly relates to orbit type be it: low earth (LEO); highly eccentric (HEO); geostationary transfer (GTO), medium earth (MEO), polar (PO), or geostationary earth (GEO). The ESA image illustrates various types of orbit, each with unique characteristics and orbital dynamics.

⁷ A person or resource which an enemy commander can least afford to lose – IRL Land Component Handbook.

⁸ A flight manoeuvre to transfer a spacecraft from one orbit to another. Markus Landgraf, ESA scientist.
<https://lunarexploration.esa.int/explore/technology/229?ia=259>



(Types of Earth Orbits, ESA 2025).

The *nature* reflects the space based activity conduct, activities are limited by launch physics, economics and space mechanics, similar to Clausewitz reflections on the *serviceability of good roads*, and opportunities presented passing through key nodes. LEO is primarily used for satellite communications (SATCOM), image intelligence (IMINT), or reconnaissance due to their proximity to earth, with MEO providing similar applications including global navigation satellite system (GNSS) at much greater cost. GEO provide SATCOM and early warning capabilities such as the U.S. DSPS⁹ or Russian EKS in HEO.



(Defence Support Program Satellite, USSF 2025).

Theoretically the *number* of LOC is vast, however giving the disposition of human activity, orbital dynamics, economic considerations, and object separation the number reduces dramatically. LOCs are becoming “*congested and competitive*” due to proliferation of satellites (MOD, 2022; USSF, 2025). Space shuttle LEO launch costs averaged €54,000 per kilogramme, with commercial enterprises the cost has reduced to €2,720 per kilogramme (Jones, 2018). European Space Agency (ESA) tracking currently estimates 1.2m objects in orbit, comprising 11,000 active satellites, 40,000

⁹ Space Force-operated Defence Support Program (DSP) satellites are a key part of North America's early warning systems.

inert objects larger than 10cm, and 1.1m objects less than 10cm. The ESA notes that the congestion of contested and competitive LOC creates chance for unexpected collisions.

The *situation* represents utility of the LOC from a military perspective, LEO reconnaissance satellites orbit every 90 minutes limiting their exposure to target areas, with electromagnetic communication from ground control stations to satellites occurring at canalisation points, additional utilities launch points for specific orbits, and common orbits all grouped.

In times of war, land, air and maritime LOC have been disrupted or blocked, in the space domain, contested LOCs float in a legal grey zone. Clausewitz requirements for protection finds a similar parallel to provide force protection on ground command and control nodes, communication and data relays and safeguarding space systems from cyber and kinetic attacks.

Engagement.

Combat is the real activity in warfare, through physical and moral effects (Clausewitz, p. 225 1832) Clausewitz argues that every military action, asset and decision are unified to support the encounter of combat, - marches, information, and sustainment are only enablers. At the tactical level, the objective of combat is destruction or conquest of enemy forces, *physically* or *morally* (Clausewitz, p. 231 1832) It requires imposing one's will on the adversary, who also is attempting to do similar actions, with equal rationality in the duel of war. The *appropriateness of means*, evaluates what methods are needed for war, requiring self comparison analysis and a logical assumption that an adversary has also conducted equal evaluations.

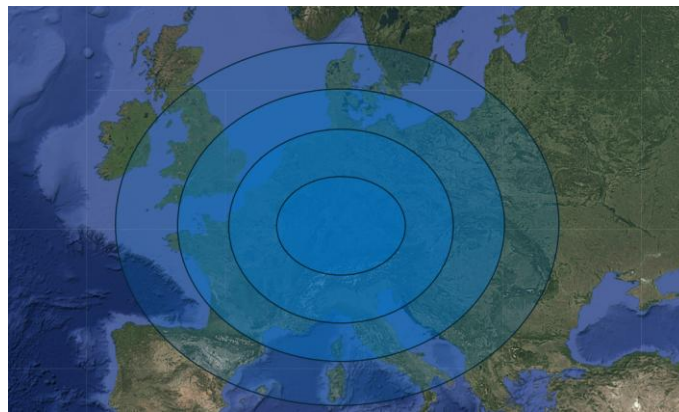
Combat cannot be accomplished within the space domain, space warfare therefore is only the means, by way of: *destruction*; *disruption*; or *denial* of adversarial space systems to achieve political objectives. A logical deduction is that the COG in space is not space systems themselves, but the overall architecture and network of information requiring the targeting on all three effects dimension¹⁰.

In campaign planning to enable operational surprise, targeting all three effects dimension requires the synchronised and continuous tactical provoking of space systems, to create mass confusion for controllers, overwhelming human OODA loop process with conflicting and false information, political signalling by the movement or readiness of strategic assets creates friction between military commanders and political leaders all to undermine confidence.

¹⁰ NATO and IRL JP-1 Joint Operations outlines three effect dimension for Joint Warfare: Physical, Virtual and Cognitive.

Kinetic means for space warfare requires direct ascent anti satellite weapons (DA-ASAT¹¹), DA-ASAT use missiles to destroy satellites, leaving high velocity debris orbiting after impact. Non-kinetic means include direct energy weapons (DEW), cyber and electronic warfare. The destruction of space systems by ground based DEW is currently not feasible (Bowen, 2021), it can minutely reduce the target velocity changing its orbital parameters (Bowen, 2021), dazzle or permanently blind satellite sensors, cause components to overheat and directed microwave transmission can permanently damage sensitive on-board electronics without causing a debris field (MOD, 2022).

Space based systems include nuclear devices to create an electromagnetic pulse (EMP), or the employment of co-orbital systems for either rendezvous¹² or proximity operations¹³. Nuclear weapons in space is prohibited by OST signatories, proxy countries¹⁴ possessing nuclear devices could use EMP as a first strike weapon, rendering unprotected electronic systems inoperable, such as modern cars, smartphones, aircraft, etc. The illustration below outlines possible EMP battle damage, if a similar device to Starfish Prime¹⁵ was detonated over Europe. Hybrid means of space warfare include manoeuvring objects to block line of sight, denying satellite capabilities, or the use of commercially register satellites to collide with friendly assets.



(EMP battle damage assessment, with similar yield and detonation characteristics to Starfish Prime, if detonate over southern Germany. McDonnell, 2025)

Critical Review

I.B. Cohen developed the concept “*the transformation of ideas*”, revealing how great scientists synthesised existing knowledge created by others, applied them as a basis or catalyst and combined

¹¹ The US, Russia, India and China have all demonstrated that they have the ability to conduct direct ascent attacks on satellites (MOD, 2022).

¹² A rendezvous operation involves attaching one satellite to another (MOD, 2022).

¹³ Proximity operations involve stationing one satellite close to another for a time (MOD, 2022).

¹⁴ A proxy country requires the means to launch a rocket into space, with a third party countries payload.

¹⁵ The Starfish Prime detonation consisted of a 1.4Mt nuclear device, deployed at 400km altitude destroying unprotected electronic equipment in Hawaii, 1450km away (Gutierrez, 2021).

them with their own works (Cohen, 1980). Sir Julian Corbett's 1911 writings *Some Principles of Maritime Strategy* were influenced by J.J. Graham's translation of *On War*, which served as his investigative source. Corbett exploited an unfinished and little understood chapter, and brilliantly interpreted it further and then further develop with his own concepts some either critiquing or complementing Clausewitz (Bellinger, 2025). Clausewitz theories on war focused on the land battle, and a states desire to avoid costly protracted wars, the ultimate demonstration being the 1991 Gulf War some 150 years after writing. Julian Corbett's writings focus on maritime warfare and strategy, which hold many similarities to space, the relevance of Clausewitz's theories is critically analysed through Corbett's optic.

Corbett reinforces some of Clausewitz writings, in particular regarding war planning and its survival with enemy encounter, Corbett's understanding of war theory informs "*what is normal, but war with its reciprocal, uncertain, and complex nature is dominated by deviation for the norm*" (Corbett, 1911). His concepts on strategic analysis also strengthen Clausewitz's thinking, *the results are not exact, only approximations of intent, requiring critical judgement*. Corbett deviates from Clausewitz, Jomini and Sun Tzu's theory regarding the decisive battle, the point of destruction for adversarial forces, by the concentration of forces (Handle, 2000). Corbett conceptualised *disposal forces* (Corbett, p.51-52 1911), significantly differing to Clausewitz's doctrine, disposal forces are aligned to modern warfighting function: *economy of force*. Disposal forces are designed around the nature of the objective, with a purpose to create strain, draw, or diverge forces otherwise required in other theatres, proven in 1809 with Wellington's expedition to Portugal. Modern operational planning utilises disposal forces as secondary efforts, or shaping activities prior, or synchronised with the main effort, these efforts enable commanders to systematically defeat adversaries, utilising the maximum effectiveness and efficiencies of available forces.

Clausewitz's *economy of force*, was not based on efficiency or return on investment, rather to make use of all available forces, regardless of the cost (Handle, 2000). Corbett's and Sun Tzu understanding of economy of force influenced modern military thinking - Iran projects power in the Strait of Hormuz and throughout the Persian Gulf using economy of force, the employment of land-based missiles, aircraft and speedboats (Brown et al., 2025). Economy of force in space could involve the deliberate positioning of low cost satellites or debris above launch facilities flight paths, the positioning of satellites along orbital canalisation points, blocking space access or severely restricting orbits. The manoeuvring of objects by an adversaries to collide with valuable military systems, citing technical malfunction would require orbital readjustment and possible loss of reconnaissance areas.

Corbett a naval strategist, articulates that concentration of naval forces is counter intuitive, that a weaker adversary will simply navigate away from such concentration, a logical assumption due to the manoeuvrability of naval forces. Corbett masterminds a ruse, to lure an adversaries attention, by displaying a fleets disposition as isolated, dispersed and vulnerable, it baits an adversary to commit larger forces, where over the horizon supporting forces wait in ambush. Corbett outlines the requirements of naval vessels to protect key points or areas, such as lines of communications, ports, or canalisation points in times of conflict. This can only be accomplished by the division of labour, assigning assets to tasks, and risk from dispersal not just for protection roles, and setting conditions to exploit relative advantages (Corbett, p. 155–61 1911). Space assets hold similar employment characteristics to maritime assets, where concentration would create vulnerabilities due to congestion of LOCs, the ability to target multiple assets from EW systems or reliance on isolated ground stations (Sheldon et Muraviev, 2021). Dispersion provides increased force protection, resilience, flexibility, operational depth and deniability of exact system capabilities.

A known criticism of Clausewitz is his undervaluing of intelligence during campaign planning. Corbett frames *terms of engagement* through the optic of intelligence led decision making, where an understanding of relative advantage, generated through an understanding of own capabilities and employment terms enables the exploitation of a decisive situation. The operation of space based systems requires an appreciation for intelligence, especially regarding adversarial capabilities, mission critical LOCs and strategic intentions.

Corbett reflects Clausewitz in repeating the importance of understanding both the value, and limitations of war theory, and its ever continuous changing nature. His divergence on concentration of forces to shape the big battle, focusing on fighting on one's own terms and luring adversaries into engagement based on deception and its relevance to modern naval warfare still holds creditability in wardrooms and bridges. The Clausewitzian concept of economy of resources fails to holds relevance in any domain of modern warfare, equipment, personnel and operations costs require efficient return on investment and therefore maximum exploitation for tactical advantage.

Conclusion

Both theorists attempted to resolve the tension between theory and practical command, and despite its growing importance, no comprehensive theory of space power has been formulated to date. Colin Gray notes that it took many centuries of experience to develop strategic theories of land and

maritime power. Clausewitz intent was to generate a field manual for the officer corps, so that the next generation would not have to relive mistakes of the past. His work gradually transformed into strategic theory and operational art, becoming mainstream education with the United States in the 1970s followed by other NATO alliance members.

Corbett's critique of Clausewitz is shaped through the maritime domain, and exposes the operational risk of over centralising strategy, particularly around LOC. Clausewitz, in the age of dispatches riders emphasised fog and friction with intelligence, where Corbett viewed intelligence as an ongoing process, albeit fluid and dynamic. Corbett writings on naval warfare provides understanding at component level, he educated decision makers, analysts on 'how to think', rather than 'what to think', he provided 'some principles', not '*The Principles*'.

Both theorists were separated by time and domain focusing on critical understandings, 21st century space warfare requires strategic understanding of theory and practical comparison with similar domains. Clausewitz and Corbett provide a planning framework and a strategic compass, aligning and integration into the full spectrum of strategic thought, operational planning for current and future multi-domain operations.

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